Lecture 25

Scattering data

The operator of Lecture 24 replaces a field by a differential equation with a spectral parameter. At each nonzero real parameter, we construct solutions with prescribed behaviour at infinity. Their comparison gives complex scattering amplitudes. Bound states require additional data.

Objectives. *We construct the Jost solutions by a convergent Volterra series. We prove the scattering flux identity with the course's Wronskian order. Translation shows which information reflection probabilities and eigenvalues lose.*

By the end of this lecture you will be able to:

- Prove convergence and both asymptotic conditions for the Jost solutions.
- Extract the scattering coefficients and prove that their denominator is nonzero.
- State the full right scattering data and the scope of the cited inverse result.

25.1 Solutions prescribed at infinity

Fix a real potential $u \in \mathcal{S}(\mathbb{R}, \mathbb{R})$, the Schwartz class of Lecture 24. Time is fixed throughout this lecture. Write $L = -\partial_x^2 + u$ and consider

$$Lf = k^2 f, \qquad k \in \mathbb{R} \setminus \{0\}.$$

The solutions may be complex. Primes denote derivatives with respect to x.

Definition 25.1 (Jost solutions)**.** A *right Jost solution* is a C^2 solution $f_+(x,k)$ with

$$f_+(x,k) - e^{\mathrm{i}kx} \to 0, \qquad f_+'(x,k) - \mathrm{i}k e^{\mathrm{i}kx} \to 0 \quad (x \to +\infty).$$

A *left Jost solution* $f_-(x,k)$ instead satisfies

$$f_-(x,k) - e^{-\mathrm{i}kx} \to 0, \qquad f_-'(x,k) + \mathrm{i}k e^{-\mathrm{i}kx} \to 0 \quad (x \to -\infty).$$

The derivative conditions are part of the normalisation. Define the *Wronskian* by $W(f,g) = fg' - f'g$. For two solutions at the same k, differentiation gives $W' = fg'' - f''g = 0$, so their Wronskian is constant.

Theorem 25.2 (Jost construction and scattering identity). *For real Schwartz u and real $k \neq 0$, both Jost solutions exist uniquely. They satisfy $f_+(x,-k) = \overline{f_+(x,k)}$. There are unique coefficients $a(k)$ and $b(k)$ such that*

$$f_-(x,k) = a(k) f_+(x,-k) + b(k) f_+(x,k). \tag{25.1}$$

With the displayed Wronskian order,

$$a(k) = \frac{W(f_-, f_+)}{2ik}, \qquad |a(k)|^2 - |b(k)|^2 = 1. \tag{25.2}$$

In particular $a(k) \neq 0$.

Proof. Volterra construction. The right integral equation is

$$f(x) = e^{ikx} + \int_x^\infty \frac{\sin k(y-x)}{k}\, u(y) f(y)\,dy. \tag{25.3}$$

To solve it, put $m(x) = e^{-ikx} f(x)$ and define

$$(Tm)(x) = \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(y) m(y)\,dy, \qquad M_u(x) = \int_x^\infty |u(y)|\,dy.$$

The kernel has absolute value at most $1/|k|$. Set $m_0 = 1$ and $m_{j+1} = T m_j$. Induction gives

$$|m_j(x)| \le \frac{M_u(x)^j}{j!\,|k|^j}. \tag{25.4}$$

Indeed, the next bound follows by integrating $|u(y)| M_u(y)^j$; since $M_u' = -|u|$, its integral from x to infinity is $M_u(x)^{j+1}/(j+1)$. Because $M_u(x) \le \|u\|_1$, the series $m = \sum_{j \ge 0} m_j$ converges uniformly on the whole line. Dominated convergence gives $m = 1 + Tm$. It is continuous and bounded by $C_k = \exp(\|u\|_1/|k|)$. Thus $f = e^{ikx} m$ solves (25.3) and is bounded by C_k.

The kernel is smooth, and uf is continuous and integrable. Differentiating the integral, with its lower endpoint included, gives

$$f'(x) = ike^{ikx} - \int_x^\infty \cos k(y-x)\, u(y) f(y)\,dy.$$

A second derivative gives

$$\begin{aligned} f''(x) &= -k^2 e^{ikx} + u(x) f(x) - k \int_x^\infty \sin k(y-x) u(y) f(y)\,dy \\ &= (u(x) - k^2) f(x). \end{aligned}$$

These differentiations follow from dominated convergence and FTC, so $f \in C^2$ and $Lf = k^2 f$. The same formulas give the actual tails

$$|f(x) - e^{ikx}| \le C_k M_u(x)/|k|, \qquad |f'(x) - ike^{ikx}| \le C_k M_u(x).$$

Both vanish as $x \to +\infty$.

Uniqueness and the left solution. Any solution with the two right asymptotic conditions satisfies the same integral equation. To see this, solve backward from a finite $R > x$:

$$f(x) = f(R)\cos k(R-x) - \frac{f'(R)}{k}\sin k(R-x) + \int_x^R \frac{\sin k(y-x)}{k}\, u(y) f(y)\,dy.$$

The endpoint expression tends to e^{ikx} as $R \to \infty$. The solution is bounded on $[x, \infty)$, so the integral converges absolutely. For two solutions, their modified difference d therefore satisfies $d = Td$. On any half-line $[x_*, \infty)$ it is bounded, and iterating the same estimate gives

$$|d(x)| \le \|d\|_{L^\infty([x_*,\infty))} \frac{M_u(x)^j}{j!|k|^j} \longrightarrow 0 \quad (j \to \infty).$$

Thus $d = 0$; the choice of x_* was arbitrary. Apply the constructed right solution to the reflected potential $u(-x)$, then replace x by $-x$. This constructs the unique left solution with both required asymptotic conditions. Conjugation and uniqueness give $f_+(x,-k) = \overline{f_+(x,k)}$, and the analogous identity for f_-.

Comparison and flux. Evaluate the constant Wronskians at the appropriate ends:

$$W(\overline{f_+}, f_+) = 2ik, \qquad W(f_-, \overline{f_-}) = 2ik.$$

The derivative tails proved above justify both limits. The first Wronskian is nonzero, so $\overline{f_+}$ and f_+ form a basis of the two-dimensional solution space. Matching a value and its first derivative at any finite point determines the two coefficients in (25.1); ordinary linear ODE uniqueness makes them independent of x. Taking $W(f_-, f_+)$ gives the formula for a. Taking the Wronskian of (25.1) with its conjugate gives

$$2ik = W(f_-, \overline{f_-}) = 2ik\big(|a|^2 - |b|^2\big).$$

Division by $2ik \neq 0$ proves (25.2) and $|a| \ge 1$. ∎

25.2 Amplitudes and the free example

Definition 25.3 (Reflection and transmission amplitudes). For real $k \neq 0$, the *right reflection amplitude* and *transmission amplitude* are

$$r(k) = b(k)/a(k), \qquad \tau(k) = 1/a(k). \tag{25.5}$$

Dividing (25.1) by a gives an incoming unit-amplitude wave e^{-ikx} at $+\infty$, a reflected wave re^{ikx} there, and a transmitted wave τe^{-ikx} at $-\infty$. Here $k > 0$ fixes that description of the incoming direction. The flux identity says $|\tau|^2 + |r|^2 = 1$. These are scattering solutions of the auxiliary operator, not finite-energy field configurations. The complex amplitude r includes phase information; its squared modulus is the reflection probability.

Example 25.4 (The free potential). For $u = 0$, the Volterra operator is zero. Thus $f_+ = e^{ikx}$ and $f_- = e^{-ikx}$. Their Wronskian is $2ik$, so $a = 1$, $b = 0$, $r = 0$ and $\tau = 1$. There are no nonzero square-integrable solutions at a negative energy: the solutions are combinations of $e^{\kappa x}$ and $e^{-\kappa x}$, and decay at both ends forces both coefficients to vanish. The nonzero travelling pulse of Lecture 23 is a different potential. Its scattering and reconstruction require the discrete data used in Lecture 26.

The proof has fixed $k \neq 0$ throughout. Its majorant contains $1/|k|$, so it gives no uniform conclusion at $k = 0$. It does not assert that a Jost construction at zero is impossible.

25.3 The discrete data and the cited inverse result

A *bound state* at energy $-\kappa^2$, with $\kappa > 0$, is a nonzero $f \in H^2(\mathbb{R})$ satisfying $Lf = -\kappa^2 f$. The space H^2 uses the Fourier norm of Lecture 24. The following spectral and inverse statements

are *cited*, not consequences of the elementary real-k theorem. For real Schwartz u, there are finitely many simple negative eigenvalues $-\kappa_j^2$. The cited Jost extension gives the real right-normalised state

$$f_j(x) = f_+(x, \mathrm{i}\kappa_j), \qquad f_j(x) \sim e^{-\kappa_j x} \ (x \to +\infty), \qquad c_j = \left(\int_{\mathbb{R}} f_j(x)^2 \mathrm{d}x \right)^{-1} > 0. \tag{25.6}$$

The asymptotic coefficient is one; an arbitrary L^2 normalisation would change c_j. Our c_j is the coefficient in the Marchenko kernel of Lecture 26 (Dunajski's β_j). For a unit-L^2 state, the magnitude of its right-tail coefficient is instead $\sqrt{c_j}$, the convention used for the constants in Drazin–Johnson. Indeed, the unit state is $f_j/\|f_j\|_2 = \sqrt{c_j} f_j$. The *full right scattering data* are

$$\big(r(k) \ (k \in \mathbb{R} \setminus \{0\}); \ \kappa_j, c_j \ (1 \le j \le N) \big).$$

Within the real Schwartz class these data determine u uniquely. This is the meaning of completeness used here; it is not a claim that arbitrary functions and positive numbers are admissible scattering data. No absence of a bounded solution at zero energy is assumed.

For a precise primary formulation, see Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), 1431–1457, author preprint, Section 4, pp. 11–16, especially (4.10), (4.26)–(4.30) and Theorems 4.3–4.4. Set both background potentials to zero. Their class (1.2) includes real Schwartz potentials, and their right norming parameter obeys $(\gamma_j^+)^2 = c_j$. In this specialisation the right integral kernel depends only on the right reflection amplitude and these discrete data. Its unique solution recovers u. The foundational reference is Deift–Trubowitz, *Communications on Pure and Applied Mathematics* **32** (1979), 121–251, doi:10.1002/cpa.3160320202. The reconstruction theorem will remain a cited input in Lecture 26.

25.4 Translation loses no energy levels but changes data

For a real shift h, let $u_h(x) = u(x - h)$. Direct substitution and the endpoint normalisations give

$$f_{+,h}(x, k) = e^{\mathrm{i}kh} f_+(x - h, k), \qquad f_{-,h}(x, k) = e^{-\mathrm{i}kh} f_-(x - h, k).$$

Substitute them into (25.1) and compare coefficients:

$$a_h = a, \qquad b_h = e^{-2\mathrm{i}kh} b, \qquad r_h = e^{-2\mathrm{i}kh} r. \tag{25.7}$$

Thus $|r_h|^2 = |r|^2$. Translation of a bound state preserves its differential equation and L^2 integrability, so the negative energy levels are unchanged. Right normalisation at the shifted potential gives

$$f_{j,h}(x) = e^{-\kappa_j h} f_j(x - h), \qquad \int f_{j,h}^2 = e^{-2\kappa_j h} \int f_j^2, \qquad c_{j,h} = e^{2\kappa_j h} c_j.$$

Any nonzero Schwartz potential differs from each nonzero translate of itself: otherwise it would be periodic with nonzero period and decay at infinity, forcing it to vanish. Consequently $|r|^2$ and the eigenvalues cannot determine the potential. The full complex reflection amplitude and the norming constants retain the positional information that those reduced data discard.

Summary. For real Schwartz potentials and real $k \neq 0$, the factorial Volterra bound constructs unique Jost solutions with both function and derivative tails. The constant Wronskian gives $|a|^2 - |b|^2 = 1$, so $r = b/a$ is defined. The full right data also contain the right-normalised bound-state constants. Their inverse uniqueness is cited with its class and conventions. Translation preserves reflection probabilities and eigenvalues. Its formulas identify the phase and norming information lost by those reduced data. The elementary proof does not cover $k = 0$.

Self-check. Why does the Volterra construction converge without a smallness condition on $\|u\|_1/|k|$?

Answer. The ordered integrations give $M_u(x)^j/(j!|k|^j)$. Its sum is an exponential, which is finite for every fixed nonzero k. A geometric-series estimate would impose an unnecessary smallness condition.

Self-check. Why is it legitimate to divide by $a(k)$? What happens to the free value of a if the Wronskian order in its formula is reversed?

Answer. The flux identity gives $|a|^2 = 1 + |b|^2 \geq 1$. For the free potential the correct Wronskian is $2ik$ and $a = 1$. Reversing its order would give $a = -1$, contrary to the fixed scattering relation.

Self-check. A right-normalised bound state is multiplied by a nonzero real number d. How does its inverse squared norm change? Is it still right-normalised?

Answer. Its inverse squared norm becomes c_j/d^2. Its coefficient of $e^{-\kappa_j x}$ becomes d, so it has the required normalisation only when $d = 1$.

Self-check. Can reflection probabilities and negative energy levels distinguish a nonzero Schwartz potential from its translate by $h \neq 0$?

Answer. No. The eigenvalues and $|r|^2$ stay fixed. The complex reflection amplitude changes by e^{-2ikh} and each right norming constant changes by $e^{2\kappa_j h}$. The omitted information is needed for inverse uniqueness.